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(54) FAST COOLING DEVICE

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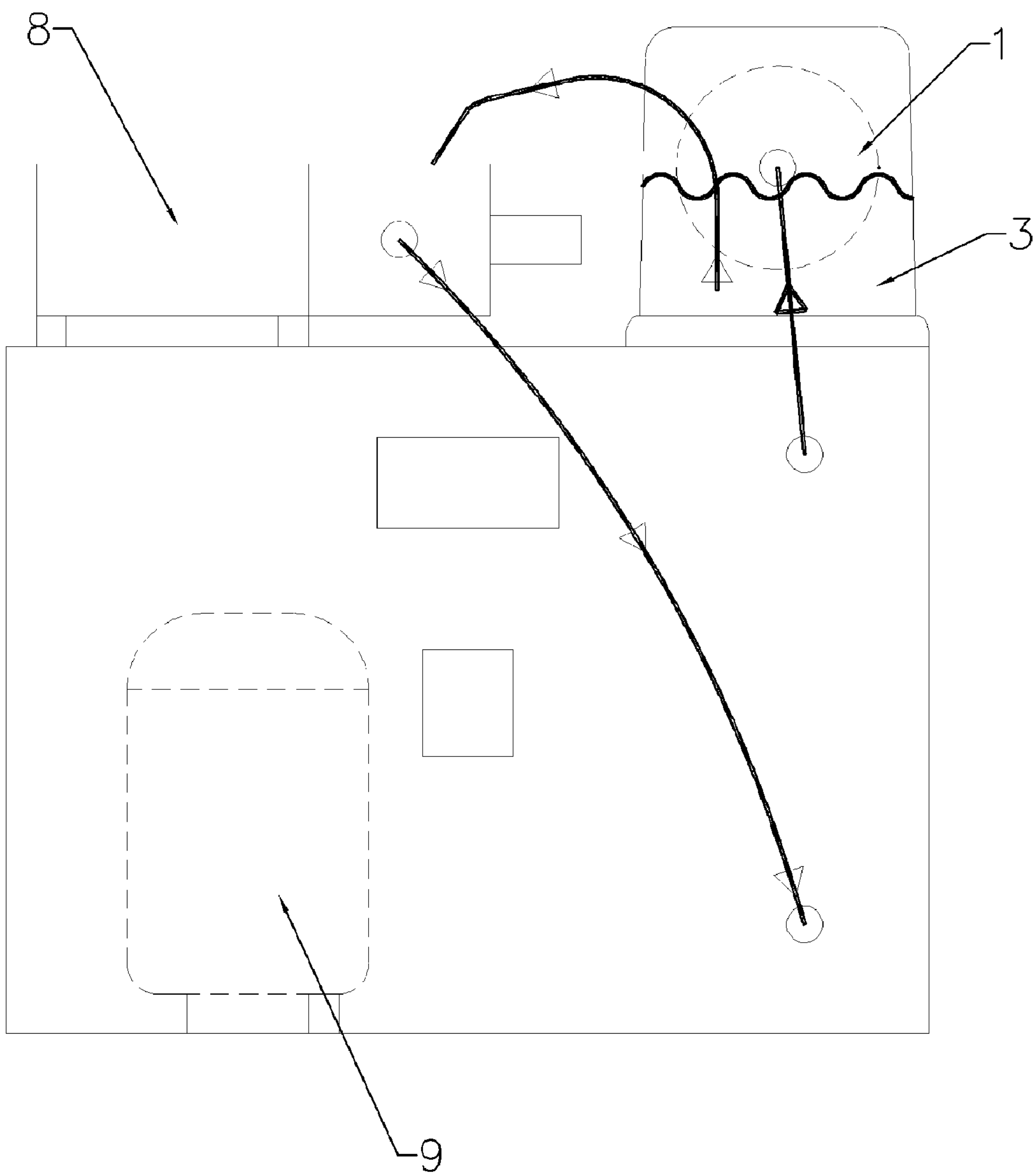
(57) **ABSTRACT**

A fast cooling device comprises a rotating device for installing and rotating the cooling container, and the rotating device is placed in the cooling space; the rotating device drives the cooling container to rotate in the cooling space. Placing the cooling container which contains liquid in the cooling space and rotating it can cause the liquid flow in the container, thereby accelerating heat conduction and convection, so that all the liquid in the cooling container can be quickly cooled down. The cooling down time reduce from one hour or more by using the existing refrigerators to a few minutes to ten minutes. Moreover, the entire device is relatively simple in structure and low production cost. It can also achieve miniaturization design and use DC power supply, so that user can quickly cool drinks conveniently anywhere.



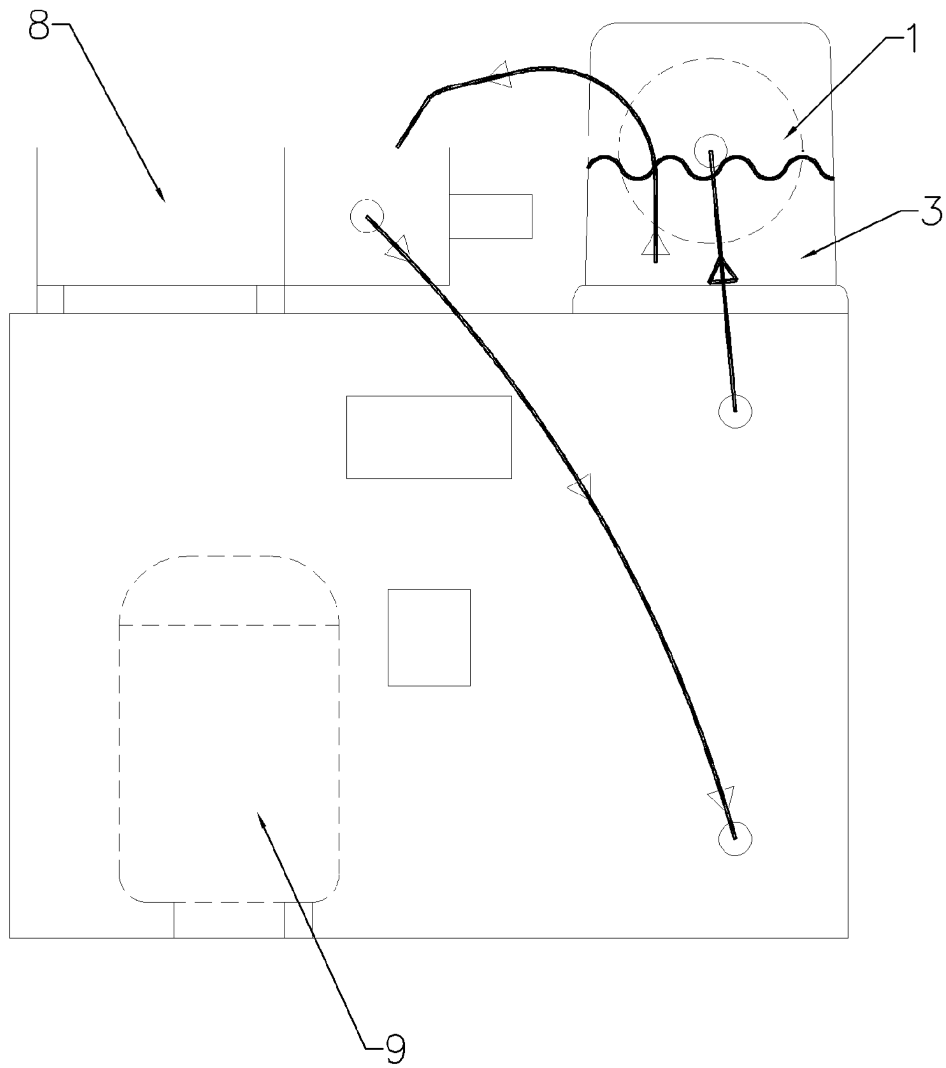


Fig. 1

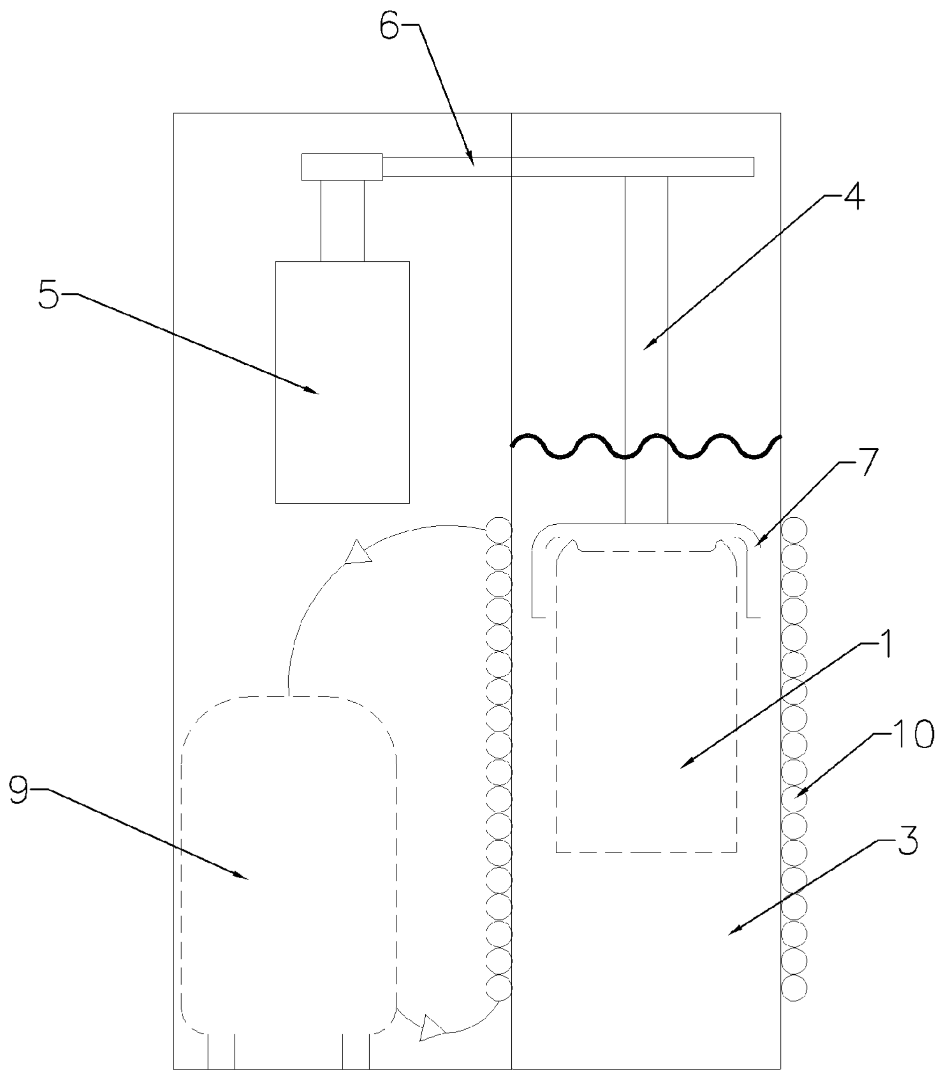


Fig. 2

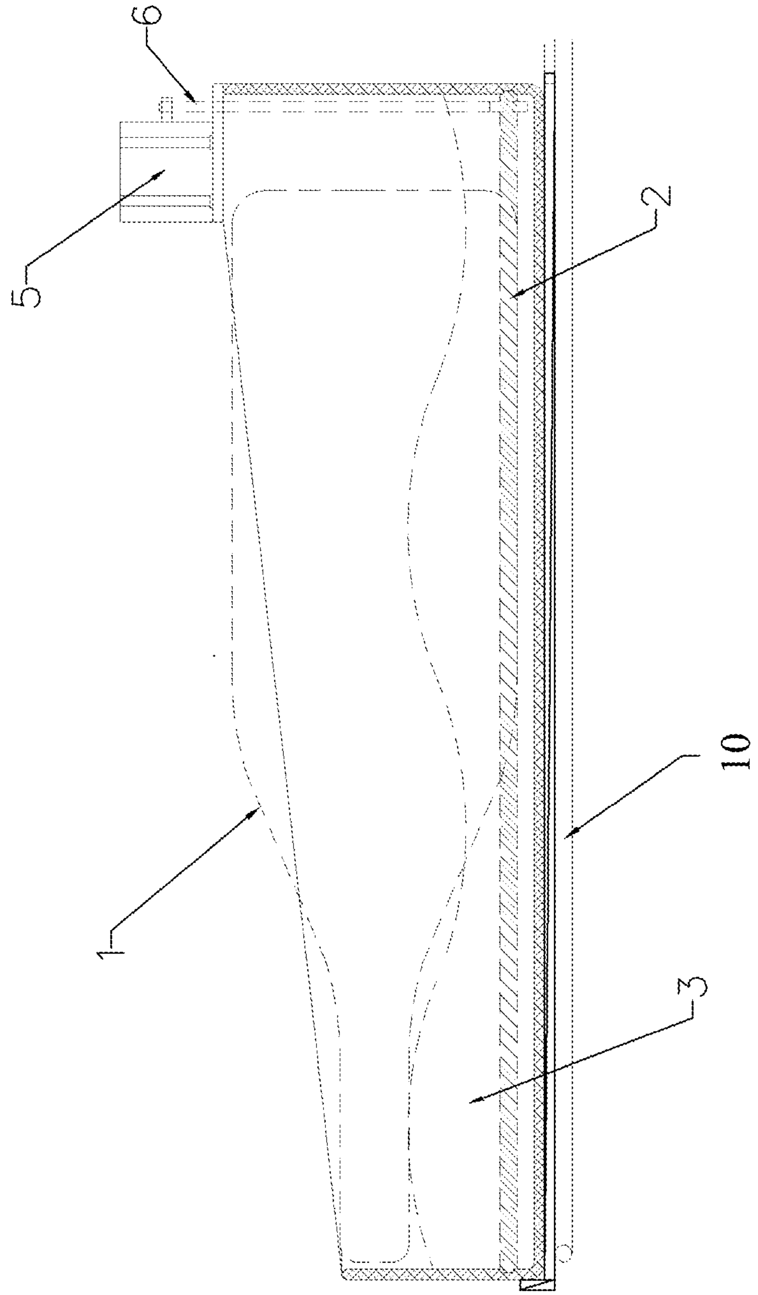


Fig. 3

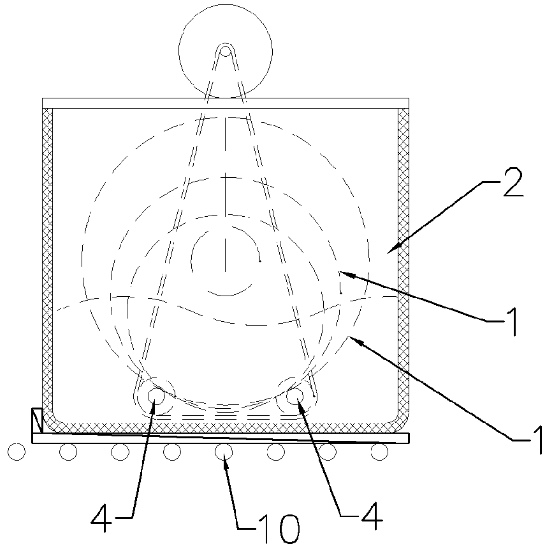


Fig. 4

FAST COOLING DEVICE

FIELD OF THE INVENTION

[0001] The invention relates to a cooling device, especially a device for fast cooling liquids in canned containers.

BACKGROUND OF THE INVENTION

[0002] In hot summer days, drinking cold drinks is a very popular way to cool down. Existing cold drinks are obtained by placing the drinks in a refrigerator to cool them down. However, this traditional method of making cold drinks mainly transfers the heat in the drink to the packaging container of the drink by thermal conduction. Due to the poor thermal conductivity of the liquid, it takes an hour or even more time to cool the drink from room temperature to about 8° C. It takes quite a long time, and if you don't put the drink in the refrigerator in advance, it won't be able to get a cold drink immediately.

SUMMARY OF THE INVENTION

[0003] In view of the aforesaid drawbacks of the prior art, the present invention provide a fast cooling device to facilitate fast cooling of canned drinks.

[0004] The invention achieves this by:

[0005] A fast cooling device, wherein comprising:

- **[0006]** a rotating device for installing and rotating the cooling container, and the rotating device is placed in the cooling space;
- **[0007]** the rotating device drives the cooling container to rotate in the cooling space.

[0008] Further, the rotating device is horizontal arranged and includes a set of parallel rotating shafts; the rotating shafts are connected to a driving device; the cooling container is placed on the rotating shafts and rotates with the rotating shafts.

[0009] Further, the rotation shafts are connected to the rotation drive device through a transmission mechanism.

[0010] Further, the transmission mechanism is a transmission belt tightened between the two rotating shafts and the output shaft of the driving device.

[0011] Further, the rotating device is vertical arranged and includes a turntable or claw plate; the cooling container is placed on the turntable or grasped and fixed by the claw plate; the turntable or claw plate is connected to the driving device.

[0012] Further, the cooling space is a container containing cooling liquid.

[0013] Further, the container is connected to a coolant circulation device, which includes a pump body and a refrigeration unit; the coolant circulates between the container and the refrigeration unit through the pump body.

[0014] Further, the cooling space is a refrigeration pipe surrounding the cooling container, and the refrigeration pipe is connected to the refrigeration unit.

[0015] The beneficial effects of the present invention are: placing the cooling container which contains liquid in the cooling space and rotating it can cause the liquid flow in the container, thereby accelerating heat conduction and convection, so that all the liquid in the cooling container can be quickly cooled down. The cooling down time reduces from one hour or more by using the existing refrigerators to a few minutes to ten minutes. Moreover, the entire device is relatively simple in structure and low production cost. It can also achieve miniaturization design and use DC power supply, so that user can quickly cool drinks conveniently anywhere.

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] The invention is further illustrated in the accompanying drawings as follows.

[0017] FIG. **1** is a schematic structural diagram of embodiment 1 of the present invention;

[0018] FIG. **2** is a schematic structural diagram of embodiment 2 of the present invention;

[0019] FIG. **3** is a front view of the rotating device and the cooling container in embodiment 1;

[0020] FIG. **4** is a right view of the rotating device and the cooling container in embodiment 1.

DESCRIPTION OF THE EMBODIMENTS

[0021] Referring to FIG. **1** and FIG. **2**, the fast cooling device comprises a rotating device **2** for installing and rotating the cooling container. The rotating device **2** is placed in the cooling space **3**. The rotating device **2** drives the cooling container **1** rotate during cooling in the cooling space **3**. By placing the cooling container **1** which contains the liquid to be cooled in the cooling space **3** and rotating it can cause the liquid flow in the container, thereby accelerating heat conduction and convection, so that all the liquid in the cooling container **1** can be rapidly cooled down. The cooling down time reduces from one hour or more by using the existing refrigerators to a few minutes to ten minutes. Moreover, the entire device is relatively simple in structure and low production cost. It can also achieve miniaturization design and use DC power supply, so that user can quickly cool drinks conveniently anywhere.

[0022] According to different usage requirements, the fast cooling device can be realized in different ways, such as horizontal and vertical arrangement, refer to FIG. **1**, FIG. **3** and FIG. **4**. For the horizontal arranged fast cooling device, refer to FIG. **3** and FIG. **4**. The rotating device **2** is horizontal arranged. Specifically it is a set of rotating shafts **4** arranged in parallel. The rotating shafts **4** are linked to the rotating driving device **5**. The cooling container **1** is placed on the rotating shaft **4** and rotates together with the rotation of the rotating shaft **4**. The advantage of using a set of parallel rotating shafts **4** is that the rotating shafts can support the cooling containers **1** of various diameters, and the outer contour of the cooling container **1** can be circular or nearly circular which can be accommodated and driven rotation by the two rotating shafts **4**, such as glass wine bottles, various beverage bottles, etc. In order to realize the driving of the rotating shaft **4**, the rotating shaft **4** is linked to a rotating driving device **5**, such as a motor, through a transmission mechanism to realize rotation. In order to simplify the entire driving mechanism, belt transmission can be used, that is, the transmission belt **6** is tightened between the two rotating shafts **4** and the output shaft of the rotating driving device **5**. The entire driving structure is very simple and easy to implement.

[0023] Refer to FIG. **2**, for a vertical arranged fast cooling device, it can be realized by a turntable or a claw plate **7**. Use the turntable as an example, the cooling container **1** can be placed directly on it. And use the claw plate **7** as an example, the top of the cooling container **1** can be grasped and fixed by the claw plate **7**. The turntable or claw plate **7** is connected to the driving device **5**, such as a motor, etc. through a transmission mechanism such as a belt or gear. The advantage of using a vertical arranged fast cooling container is that it can be applied to containers with various outer contours and does not require a round or quasi-round tank.

[0024] Similarly, the cooling space **3** can also be implemented using a variety of different structures or cooling methods according to actual needs. If the coolant is used as one of the cooling methods, more specifically the cooling space **3** is set as a container filled with coolant. In order to maintain the low temperature of the coolant, the container is connected to a coolant circulation device, which includes a pump body **8** and a refrigeration unit **9**. The coolant circulates between the container and the refrigeration unit **9** through the pump body **8**, so that the coolant can be kept at a low temperature, and the cooling container **1** can be continuously replaced and cooled. Another cooling method is to use the refrigeration pipe **10**, which is arranged around the cooling container **1**. The refrigeration pipe **10** is connected to the refrigeration unit **9** for continuous refrigeration. The above refrigeration unit **9** can adopt electronic refrigeration or compression refrigeration, etc.

[0025] The above are only preferred embodiments of the present invention. The present invention is not limited to the above-mentioned embodiments. As long as the technical effects of the present invention are achieved by the similar methods and everything within the spirit and principles of the present invention, any modifications, equivalent substitutions, improvements, etc. shall be included in the scope of protection of this disclosure. All belong to the protection scope of the present invention. Various modifications and changes may be made to the technical solutions and/or implementations within the protection scope of the present invention.

1. A fast cooling device, wherein comprising:

a rotating device for installing and rotating the cooling container, and the rotating device is placed in the cooling space;

the rotating device drives the cooling container to rotate in the cooling space.

2. The fast cooling device of claim **1**, wherein the rotating device is horizontal arranged and includes a set of parallel rotating shafts; the rotating shafts are connected to a driving device;

the cooling container is placed on the rotating shafts and rotates with the rotating shafts.

3. The fast cooling device of claim **2**, wherein the rotation shafts are connected to the rotation drive device through a transmission mechanism.

4. The fast cooling device of claim **3**, wherein the transmission mechanism is a transmission belt tightened between the two rotating shafts and the output shaft of the driving device.

5. The fast cooling device of claim **1**, wherein the rotating device is vertical arranged and includes a turntable or claw plate; the cooling container is placed on the turntable or grasped and fixed by the claw plate; the turntable or claw plate is connected to the driving device.

6. The fast cooling device of claim **1**, wherein the cooling space is a container containing cooling liquid

7. The fast cooling device of claim **6**, wherein the container is connected to a coolant circulation device, which includes a pump body and a refrigeration unit; the coolant circulates between the container and the refrigeration unit through the pump body.

8. The fast cooling device of claim **1**, wherein the cooling space is a refrigeration pipe surrounding the cooling container, and the refrigeration pipe is connected to the refrigeration unit.

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